# LAB 06

## SUBMISSION INSTRUCTIONS

Type/write your answers on the document and submit it as a pdf file with the name JaneDoe6.pdf (replace JaneDoe with your first and last name respectively).

## QUESTIONS

1. (10pts): What is the simplified Big O notation of the expressions below:
   a. O(n/2 + 99) : O(n)

   b. O((n$^3$ + 3) * (O(n) + 5)) : O(n^4)

   c. O(12n +6n$^3$ + 1000) : O(n^3)

   d. 2n$^3$+ O(n$^2$): O(n^3)

   e. $\log_2 n + 2 + \log_5 n$ : O(logn)

2. (20pts): What is the time complexity of the functions below (express answer in simplified Big O notation)?

| | Code | Time complexity |
|---|---|---|
| a) | `def test(n):`<br>`  for i in range(n):`<br>`  for j in range(n):`<br>`  print(i, j)` | O(n^2) |
| b) | `name = ['Jane', 'John', 'Jill']`<br>`def check(names):`<br>`  for value in names:`<br>`  if value == 'John':`<br>`  print('Found')`<br>`check(name)` | O(n) |
| c) | `def test(x, y):`<br>`  for i in range(x):`<br>`  print(i)`<br><br>`  for j in range(y):`<br>`  print(j)` | O(x+y) |
| d) | `def test(n):`<br>`  i = n`<br>`  while i > 0:`<br>`  i = i - 2`<br>`  print(i)` | O(n) |

| | | |
|---|---|---|
| e) | <br>`def test(n, m):`<br>`  for i in range(m):`<br>`  for j in range(m):`<br>`  k = n`<br>`  while k > 0:`<br>`  k = k - 2`<br>`  print(i, j, k)` | O(m^2*n) |
| f) | <br>`name = ['Jane', 'John', 'Jill']`<br>`def display(names):`<br>`  print(names[0])`<br>`  print(names[2])`<br>`display(name)` | O(1) |
| g) | <br>`def test(n):`<br>`  for i in range(n):`<br>`  print(i)`<br><br>`  for i in range(n):`<br>`  for j in range(n):`<br>`  for k in range(n):`<br>`  print(i, j, k)` | O(n^3) |
| h) | <br>`def test(old_list):`<br>`  new_list = []`<br>`  for value in old_list:`<br>`  new_list.append(value)`<br>`  return new_list` | O(n) |
| i) | <br>`def test(n):`<br>`  for i in range(0, n):`<br>`  k = 1`<br>`  while k < n:`<br>`  k = k * 2`<br>`  print(i, k)` | O(nlogn) |
| j) | <br>`def test(n):`<br>`  for i in range(0, n):`<br>`  k = 1`<br>`  while k < n:`<br>`  k = k * 2`<br>`  print(i, k)`<br><br>`  for i in range(0, n):`<br>`  for j in range(0, n):`<br>`  for k in range(0, n):`<br>`  print(i, j, k)` | O(n^3) |